

Obesity_Indicators_Filtering

December 11, 2025

```
[44]: import pandas as pd
```

```
# Load all available indicators  
df = pd.read_csv('../data/who_obesity_data.csv')  
df.shape
```

C:\Users\Jack\AppData\Local\Temp\ipykernel_21604\4030832271.py:5: DtypeWarning: Columns (0,1,2,4,5,6,7,8,9,17,18,25,26,27) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv('../data/who_obesity_data.csv')
```

```
[44]: (3961866, 28)
```

```
[63]: # Calculate total number of unique sub-dimensions in Dim1, Dim2, and Dim3
```

```
subdim_cols = ['Dim1', 'Dim2', 'Dim3']  
unique_subdims = set()  
for col in subdim_cols:  
    unique_subdims.update(df[col].dropna().unique())  
print(f"Total unique sub-dimensions: {len(unique_subdims)}")
```

```
# Calculate total number of unique sub-dimensions in Dim2 only  
num_unique_dim2 = df['Dim2'].dropna().nunique()  
print(f"Total unique sub-dimensions in Dim2: {num_unique_dim2}")
```

Total unique sub-dimensions: 5303

Total unique sub-dimensions in Dim2: 119

```
[55]: # Examine dimension values
```

```
print("Dim1 (Sex) Value counts:")  
print(df['Dim1'].value_counts(dropna=False))  
print("\n" + "="*50 + "\n")
```

```
print("Dim2 (Categories) - Value counts:")  
print(df['Dim2'].value_counts(dropna=False))  
print("\n" + "="*50 + "\n")
```

```
print("Dim1Type values:")
```

```
print(df['Dim1Type'].value_counts(dropna=False))
print("\n" + "="*50 + "\n")

print("Dim2Type values:")
print(df['Dim2Type'].value_counts(dropna=False))
```

Dim1 (Sex) Value counts:

```
Dim1
SEX_FMLE          907147
SEX_MLE           905115
SEX_BTSX          602371
NaN               394149
RESIDENCEAREATYPE_TOTL 131667
...
AGEGROUP_YEARS30-39      2
AGEGROUP_YEARS10-14      2
AGEGROUP_YEARS30-49      2
AGEGROUP_YEARS10-19      2
AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS07 2
Name: count, Length: 5070, dtype: int64
```

=====

Dim2 (Categories) - Value counts:

```
Dim2
NaN          1946809
AGEGROUP_YEARS18-PLUS 305859
AGEGROUP_AGEAll      136244
AGEGROUP_YEARS10-19  113724
AGEGROUP_YEARS15-29   89661
...
ASSISTIVETECHSOURCE_ASSISTIVETECH_Private 52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Self-made 52
ASSISTIVETECHSOURCE_ASSISTIVETECH_NGO      52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Public   52
EMFFREQUENCY_EMFSTATIC 49
Name: count, Length: 120, dtype: int64
```

=====

Dim1Type values:

```
Dim1Type
SEX          2414633
NaN          394149
RESIDENCEAREATYPE 385514
TOBACCO_INDICATOR 376244
ALCOHOLTYPE    138284
DHSMICSGEOREGION 40410
```

WEALTHDECILE	40351
AGEGROUP	29580
WEALTHQUINTILE	25215
EDUCATIONLEVEL	18047
SEVERITY	16160
TOBACCO_NICOTINE_PRODUCT	13065
RSUDREP	7760
HOUSEHOLD_COMP_BY_AGE	6902
HOUSEHOLDWEALTH	5518
RSUDSPECIFICPOPULATION	5432
RSUDTREATMENTLOCATION	4656
RSUDHWF	3880
BEVERAGETYPE	3854
SCSUDSURVEYQUESTIONS	3335
PUBLICPLACE	2304
PROGRAMME	1940
TB_TREATHISTORY	1899
DRUG	1882
ADVERTISINGTYPE	1626
ENVCAUSE	1552
RSUDPREVENTIONPROGRAMME	1552
SPECIALPOPULATION	1305
POLICYSECTOR	1237
GOEQUESTION	1223
ARCHIVE	1153
WEALTHTERCILE	1048
BACGROUP	1044
SUBSTANCETYPEDISORDER	1022
GROUP	970
NATIONALSYSTEMTYPE	878
COORDINATINGENTITIES	840
RSUDTREATMENT	776
OPENACCESSSERVICE	582
NGO	544
GOVERNMENTBENEFIT	498
POLICYFRAMEWORK	498
ASSISTIVETECHSUBQUESTION	490
RSUDMONITORING	388
DRIVERTYPE	350
ASSISTIVETECHPRODUCT	312
HARMANDCONSEQUENCE	275
EMFEXPOSED	232
CONSUMPTIONTYPE	126
SOCIALCOSTTYPE	108
TB_TREATMENTTYPE	92
LEGISLATIONTYPE	55
POLICYADOPTIONLEVEL	47
AMRGLASSCATEGORY	24

PRICE 5
Name: count, dtype: int64

=====

Dim2Type values:
Dim2Type
NaN 1946809
AGEGROUP 1582086
HOUSEHOLDCOOKINGFUEL 204408
GHECAUSE 135720
TOBACCO_NICOTINE_PRODUCT 36855
ENVCAUSE 16722
ARCHIVE 16652
PRICE 12675
POP_TYPE 6658
BEVERAGE 2340
TB_CASETYPE 397
ASSISTIVETECHSOURCE 312
EMFFREQUENCY 232
Name: count, dtype: int64

```
[56]: # Check all Dim1Type categories and their top values
print("=" * 60)
print("DIM1 TYPES AND THEIR VALUES")
print("=" * 60)

for dim_type in df['Dim1Type'].dropna().unique():
    dim_values = df[df['Dim1Type'] == dim_type]['Dim1'].value_counts().head(10)
    print(f"\n{dim_type}:")
    print(dim_values)
    print("-" * 40)

print("\n\n")
print("=" * 60)
print("DIM2 TYPES AND THEIR VALUES")
print("=" * 60)

for dim_type in df['Dim2Type'].dropna().unique():
    dim_values = df[df['Dim2Type'] == dim_type]['Dim2'].value_counts().head(10)
    print(f"\n{dim_type}:")
    print(dim_values)
    print("-" * 40)
```

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DIM1 TYPES AND THEIR VALUES

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AMRGLASSCATEGORY:

Dim1

AMRGLASSCATEGORY_AMRGLASS_UNKNOWN	12
AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS06	10
AMRGLASSCATEGORY_AMRGLASS_PROVIDEDTOLABS07	2

Name: count, dtype: int64

ASSISTIVETECHSUBQUESTION:

Dim1

ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_02	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_06	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_04	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_01	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_05	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_Q07_03	70
ASSISTIVETECHSUBQUESTION_ASSISTIVETECH_SUMMARY	70

Name: count, dtype: int64

ASSISTIVETECHPRODUCT:

Dim1

ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL	156
ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL_NOSPECS	156

Name: count, dtype: int64

SEX:

Dim1

SEX_FMLE	907147
SEX_MLE	905115
SEX_BTSEX	602371

Name: count, dtype: int64

EMFEXPOSED:

Dim1

EMFEXPOSED_EMFWORKER	117
EMFEXPOSED_EMFPUBLIC	115

Name: count, dtype: int64

WEALTHQUINTILE:

Dim1

WEALTHQUINTILE_WQ3	4585
WEALTHQUINTILE_WQ2	4585
WEALTHQUINTILE_WQ1	4585
WEALTHQUINTILE_WQ4	4585

WEALTHQUINTILE_WQ5 4584
 WEALTHQUINTILE_TOTL 2291
 Name: count, dtype: int64

RESIDENCEAREATYPE:

Dim1
 RESIDENCEAREATYPE_TOTL 131667
 RESIDENCEAREATYPE_URB 127578
 RESIDENCEAREATYPE_RUR 126269
 Name: count, dtype: int64

HOUSEHOLD_COMP_BY_AGE:

Dim1
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_MULTIGEN 1184
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS_WITH_KIDS 1151
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS 1129
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_ADULTS_WITH_OLDER 1124
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_MULTIGEN_WITH_OLDER 1105
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_ONLY_OLDER_ADULTS 1062
 HOUSEHOLD_COMP_BY_AGE_HH_COMP_ONLY_BELOW_20_YEARS 147
 Name: count, dtype: int64

AGEGROUP:

Dim1
 AGEGROUP_YEARS20-49 3456
 AGEGROUP_YEARS15-19 3456
 AGEGROUP_YEARSALL 1668
 AGEGROUP_AGE60+ 1451
 AGEGROUP_AGE16-60 1451
 AGEGROUP_MONTHS6-11 990
 AGEGROUP_MONTHS12-23 990
 AGEGROUP_MONTHS24-35 765
 AGEGROUP_MONTHS36-47 759
 AGEGROUP_MONTHS48-59 751
 Name: count, dtype: int64

GOEQUESTION:

Dim1
 GOEQUESTION_GOEQUESTION016 366
 GOEQUESTION_GOEQUESTION014 242
 GOEQUESTION_GOEQUESTION009 123
 GOEQUESTION_GOEQUESTION013 123
 GOEQUESTION_GOEQUESTION011 123
 GOEQUESTION_GOEQUESTION012 123

GOEQUESTION_GOEQUESTION015 123

Name: count, dtype: int64

TOBACCO_INDICATOR:

Dim1

TOBACCO_INDICATOR_PRICE_IN_CURRENCY	3705
TOBACCO_INDICATOR_PRICE_IN_PPP	3705
TOBACCO_INDICATOR_CURRENCY_LOCAL	3705
TOBACCO_INDICATOR_PRICE_IN_USD	3705
TOBACCO_INDICATOR_TAX_EXCISEPROPPRICE	2925
TOBACCO_INDICATOR_TAX_VALUEADDED	2925
TOBACCO_INDICATOR_TAX_IMPORTDUTIES	2925
TOBACCO_INDICATOR_TAX_TOTALTAX	2925
TOBACCO_INDICATOR_TAX_SPECIFICEXCISE	2925
TOBACCO_INDICATOR_TAX_ADVALEXCISE	2925

Name: count, dtype: int64

TOBACCO_NICOTINE_PRODUCT:

Dim1

TOBACCO_NICOTINE_PRODUCT_CIGARETTE_MOSTSOLD	6240
TOBACCO_NICOTINE_PRODUCT_TOBACCO_OTHERSMOKED_MOSTSOLD	1170
TOBACCO_NICOTINE_PRODUCT_TOBACCO_SMOKELESS_MOSTSOLD	975
TOBACCO_NICOTINE_PRODUCT_CIGARETTE_PREMIUM	780
TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_RECHARGABLE_CHEAPEST	780
TOBACCO_NICOTINE_PRODUCT_TOBACCO_HEATED_HTP_MOSTSOLD	780
TOBACCO_NICOTINE_PRODUCT_CIGARETTE_CHEAPEST	780
TOBACCO_NICOTINE_PRODUCT_ELILQUID_OPEN_CHEAPEST	780
TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_DISPOSABLE_CHEAPEST	780

Name: count, dtype: int64

PRICE:

Dim1

PRICE_PRICE_IN_PPP 5

Name: count, dtype: int64

DHSMICSGEOREGION:

Dim1

DHSMICSGEOREGION_KHMDHS200014	9
DHSMICSGEOREGION_PERDHS200413	9
DHSMICSGEOREGION_RWADHS201002	9
DHSMICSGEOREGION_GMBDHS201902	9
DHSMICSGEOREGION_PERDHS200913	9
DHSMICSGEOREGION_NAMDHS200004	9
DHSMICSGEOREGION_IRQMICS201805	9

DHSMICSGEOREGION_SENDHS201612	9
DHSMICSGEOREGION_VUTMICS200705	9
DHSMICSGEOREGION_CODDHS201304	9

Name: count, dtype: int64

WEALTHDECILE:

Dim1	
WEALTHDECILE_WEALTHDECILE09	3932
WEALTHDECILE_WEALTHDECILE08	3932
WEALTHDECILE_WEALTHDECILE03	3932
WEALTHDECILE_WEALTHDECILE10	3932
WEALTHDECILE_WEALTHDECILE06	3932
WEALTHDECILE_WEALTHDECILE02	3932
WEALTHDECILE_WEALTHDECILE01	3931
WEALTHDECILE_WEALTHDECILE04	3931
WEALTHDECILE_WEALTHDECILE07	3931
WEALTHDECILE_WEALTHDECILE05	3930

Name: count, dtype: int64

EDUCATIONLEVEL:

Dim1	
EDUCATIONLEVEL_SHLM	4552
EDUCATIONLEVEL_PRLM	4498
EDUCATIONLEVEL_NHLM	4439
EDUCATIONLEVEL_TOTL	1271
EDUCATIONLEVEL_EDUC_NONE_PRIMARY	632
EDUCATIONLEVEL_EDUC_SECONDARY	596
EDUCATIONLEVEL_EDUC_HIGHER	507
EDUCATIONLEVEL_MASTER	388
EDUCATIONLEVEL_BACHELOR	388
EDUCATIONLEVEL_DOCTORAL	388

Name: count, dtype: int64

SEVERITY:

Dim1	
SEVERITY_TOTAL	4040
SEVERITY_MILD	4040
SEVERITY_MODERATE	4040
SEVERITY_SEVERE	4040

Name: count, dtype: int64

HOUSEHOLDWEALTH:

Dim1	
HOUSEHOLDWEALTH_TOTL	1271

HOUSEHOLDWEALTH_WEALTH_TOP40	638
HOUSEHOLDWEALTH_WEALTH_TOP60	638
HOUSEHOLDWEALTH_WEALTH_BOTTOM60	638
HOUSEHOLDWEALTH_WEALTH_BOTTOM80	638
HOUSEHOLDWEALTH_WEALTH_TOP80	638
HOUSEHOLDWEALTH_WEALTH_BOTTOM40	638
HOUSEHOLDWEALTH_WEALTH_MIDDLE60	419

Name: count, dtype: int64

WEALTHTERCILE:

Dim1	
WEALTHTERCILE_TOTL	1036
WEALTHTERCILE_WEALTHTERCILE2	4
WEALTHTERCILE_WEALTHTERCILE1	4
WEALTHTERCILE_WEALTHTERCILE3	4

Name: count, dtype: int64

SUBSTANCETYPEDISORDER:

Dim1	
SUBSTANCETYPEDISORDER_ALCOHOL_USE_DISORDERS	512
SUBSTANCETYPEDISORDER_DRUG_USE_DISORDERS	510

Name: count, dtype: int64

RSUDTREATMENTLOCATION:

Dim1	
RSUDTREATMENTLOCATION_IN_THE_CAPITAL_CITY	1552
RSUDTREATMENTLOCATION_IN_OTHER_AREAS	1552
RSUDTREATMENTLOCATION_IN_OTHER_MAJOR_CITIES	1552

Name: count, dtype: int64

GROUP:

Dim1	
GROUP_AL_ANON_ALATEEN	194
GROUP_COCAINE_ANONYMOUS	194
GROUP_FAMILY_SUPPORT_GROUPS	194
GROUP_NARCOTICS_ANONYMOUS	194
GROUP_ALCOHOLICS_ANONYMOUS	194

Name: count, dtype: int64

LEGISLATIONTYPE:

Dim1	
LEGISLATIONTYPE_SA_ILLEGAL_PRODUCTION	45
LEGISLATIONTYPE_SA_ILLEGAL_SALES	10

Name: count, dtype: int64

ADVERTISINGTYPE:

Dim1

ADVERTISINGTYPE_SA_WINE_ADS	542
ADVERTISINGTYPE_SA_BEER_ADS	542
ADVERTISINGTYPE_SA_SPIRITS_ADS	542

Name: count, dtype: int64

BEVERAGETYPE:

Dim1

BEVERAGETYPE_ALCOHOLIC	1950
BEVERAGETYPE_NONALCOHOLIC	390
BEVERAGETYPE_SA_750_POPULAR_IMPORTED_SPIRITS	380
BEVERAGETYPE_SA_500_POPULAR_BEER	378
BEVERAGETYPE_SA_750_TABLE_WINE	378
BEVERAGETYPE_SA_750_POPULAR_LOCAL_SPIRITS	378

Name: count, dtype: int64

ALCOHOLTYPE:

Dim1

ALCOHOLTYPE_SA_BEER	31196
ALCOHOLTYPE_SA_SPIRITS	31191
ALCOHOLTYPE_SA_WINE	31188
ALCOHOLTYPE_SA_TOTAL	22754
ALCOHOLTYPE_SA_OTHER_ALCOHOL	21955

Name: count, dtype: int64

SCSUDSURVEYQUESTIONS:

Dim1

SCSUDSURVEYQUESTIONS_SCSUD41_07_independentbody_licensing_health_profs	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_community_hlthwrkr	145
SCSUDSURVEYQUESTIONS_SCSUD41_06_attain_masters_or_equivalent_level	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_field_workers	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_nurses	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_nurses	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_psychiatrists	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_postgradtrainprogs_addict_counsellors	145
SCSUDSURVEYQUESTIONS_SCSUD41_05_contprofdev_narcologists	145
SCSUDSURVEYQUESTIONS_SCSUD41_06_attain_short_cycle_tertiary_education	145

Name: count, dtype: int64

TB_TREATMENTTYPE:

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Dim1
TB_TREATMENTTYPE_TB_TREATMENTTYPE_DRTB      36
TB_TREATMENTTYPE_TB_TREATMENTTYPE_DSTB      36
TB_TREATMENTTYPE_TB_TREATMENTTYPE_TOTL      20
Name: count, dtype: int64
-----

TB_TREATHISTORY:
Dim1
TB_TREATHISTORY_TB_TREATHISTORY_NEW          959
TB_TREATHISTORY_TB_TREATHISTORY_RETREATED    940
Name: count, dtype: int64
-----

DRIVERTYPE:
Dim1
DRIVERTYPE_RS-DRIVERTYPE-YOUNG               175
DRIVERTYPE_RS-DRIVERTYPE-GENERAL             175
Name: count, dtype: int64
-----

RSUDREP:
Dim1
RSUDREP_RSUDREP_NGO                          776
RSUDREP_RSUDREP_TOBACCOINDUSTRY              776
RSUDREP_RSUDREP_ALCOHOLUSEDISORDERFAM        776
RSUDREP_RSUDREP_DRUGUSER                     776
RSUDREP_RSUDREP_DRUGUSEDISORDER              776
RSUDREP_RSUDREP_ALCOHOLINDUSTRY              776
RSUDREP_RSUDREP_PHARMAINDUSTRY               776
RSUDREP_RSUDREP_DRUGUSEDISORDERFAM           776
RSUDREP_RSUDREP_YOUTH                        776
RSUDREP_RSUDREP_ALCOHOLUSEDISORDER           776
Name: count, dtype: int64
-----

RSUDTREATMENT:
Dim1
RSUDTREATMENT_RSUDTREATMENT_ALTERNATIVE      388
RSUDTREATMENT_RSUDTREATMENT_ADDITION          388
Name: count, dtype: int64
-----

GOVERNMENTBENEFIT:
Dim1
GOVERNMENTBENEFIT_NON_MONETARY_SUPPORT        194
GOVERNMENTBENEFIT_MONETARY_SUPPORT            194
GOVERNMENTBENEFIT_DISABILITY_PENSION          55

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GOVERNMENTBENEFIT_SUBSIDY_FOOD_HOUSING 55

Name: count, dtype: int64

OPENACCESSSERVICE:

Dim1

OPENACCESSSERVICE_WEB_BASED_INTERVENTIONS 194

OPENACCESSSERVICE_MOBILE_PHONE_BASED_INTERVENTIONS 194

OPENACCESSSERVICE_TELEPHONE_HELP_LINES 194

Name: count, dtype: int64

RSUDHWF:

Dim1

RSUDHWF_ADDICTION_COUNSELLORS 388

RSUDHWF_DOCTORS_NOT_SPECIALIZED_IN_PSYCH_OR_ADDICTMED 388

RSUDHWF_PSYCHIATRIC_NURSES 388

RSUDHWF_SOCIAL_WORKERS 388

RSUDHWF_OUTREACH_FIELD_WORKERS 388

RSUDHWF_PSYCHIATRISTS 388

RSUDHWF_COMMUNITY_HEALTH_WORKERS 388

RSUDHWF_NURSES_NOT_SPECIALIZED_IN_PSYCH 388

RSUDHWF_NARCOLOGISTS 388

RSUDHWF_PSYCHOLOGISTS 388

Name: count, dtype: int64

PROGRAMME:

Dim1

PROGRAMME_IN_PATIENT_TREATMENT 388

PROGRAMME_OUT_PATIENT_TREATMENT 388

PROGRAMME_OUT_PATIENT_DETOXIFICATION 388

PROGRAMME_REHABILITATION 388

PROGRAMME_IN_PATIENT_DETOXIFICATION 388

Name: count, dtype: int64

NGO:

Dim1

NGO_PREVENTION 137

NGO_REHABILITATION 136

NGO_TREATMENT 136

NGO_ADVOCACY 135

Name: count, dtype: int64

RSUDMONITORING:

Dim1

```

RSUDMONITORING_FORENSIC_EXAMS      194
RSUDMONITORING_TOXICOLOGY_UNITS    194
Name: count, dtype: int64
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DRUG:

```

Dim1
DRUG_ACAMPROSATE      318
DRUG_DISULFIRAM      194
DRUG_NALTREXONE      194
DRUG_DIAZEPAM        194
DRUG_BENZODIAZEPINES 131
DRUG_CHLORPROMAZINE  126
DRUG_NEW_ANTIPSYCHOTICS 125
DRUG_ALCOHOL_INFUSION 121
DRUG_TIAGABINE       120
DRUG_NITROUS_OXIDE   120
Name: count, dtype: int64
-----

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SPECIALPOPULATION:

```

Dim1
SPECIALPOPULATION_CHILDREN_AND_FAMILIES_RISK      145
SPECIALPOPULATION_CHILDREN_AND_ADOLESCENTS      145
SPECIALPOPULATION_COMMERCIAL_SEX_WORKERS      145
SPECIALPOPULATION_REFUGEES      145
SPECIALPOPULATION_PEOPLE_LIVING_WITH_HIV_AIDS      145
SPECIALPOPULATION_MINORITY_GROUPS      145
SPECIALPOPULATION_PREGNANT_WOMEN      145
SPECIALPOPULATION_PRISONERS      145
SPECIALPOPULATION_INDIGENOUS_POPULATIONS      145
Name: count, dtype: int64
-----

```

RSUDSPECIFICPOPULATION:

```

Dim1
RSUDSPECIFICPOPULATION_CHILDREN_AND_FAMILIES_RISK      388
RSUDSPECIFICPOPULATION_REFUGEES      388
RSUDSPECIFICPOPULATION_COMMERCIAL_SEX_WORKERS      388
RSUDSPECIFICPOPULATION_CHILDREN_AND_ADOLESCENTS      388
RSUDSPECIFICPOPULATION_YOUNG_ADULTS      388
RSUDSPECIFICPOPULATION_PARENTS      388
RSUDSPECIFICPOPULATION_INDIGENOUS_POPULATIONS      388
RSUDSPECIFICPOPULATION_PERSONS_WITH_MENTAL_HEALTH_DISORDERS 388
RSUDSPECIFICPOPULATION_PREGNANT_WOMEN      388
RSUDSPECIFICPOPULATION_PEOPLE_WITH_TB      388
Name: count, dtype: int64
-----

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RSUDPREVENTIONPROGRAMME:

Dim1

RSUDPREVENTIONPROGRAMME_MASS_MEDIA_PRINT	194
RSUDPREVENTIONPROGRAMME_SCHOOL_MANUAL_BASED	194
RSUDPREVENTIONPROGRAMME_ADS_IN_PUBLIC	194
RSUDPREVENTIONPROGRAMME_MASS_MEDIA_AUDIOVISUAL	194
RSUDPREVENTIONPROGRAMME_WORK_PLACE_PROG	194
RSUDPREVENTIONPROGRAMME_COMMUNITY_BASED_PROG	194
RSUDPREVENTIONPROGRAMME_PARENT_PROG	194
RSUDPREVENTIONPROGRAMME_SERVING_STAFF_TRAINING_PROG	194

Name: count, dtype: int64

CONSUMPTIONTYPE:

Dim1

CONSUMPTIONTYPE_CONSUMPTION_UNRECORDED	42
CONSUMPTIONTYPE_CONSUMPTION_RECORDED	42
CONSUMPTIONTYPE_CONSUMPTION_TOTAL	42

Name: count, dtype: int64

ARCHIVE:

Dim1

ARCHIVE_DEC2019 1153

Name: count, dtype: int64

SOCIALCOSTTYPE:

Dim1

SOCIALCOSTTYPE_SA_DIRECT_OTHER_COSTS	22
SOCIALCOSTTYPE_SA_INDIRECT_COSTS	22
SOCIALCOSTTYPE_SA_DIRECT_HEALTH_COSTS	22
SOCIALCOSTTYPE_SA_TOTAL_COSTS	22
SOCIALCOSTTYPE_SA_DIRECT_LAW_COSTS	20

Name: count, dtype: int64

HARMANDCONSEQUENCE:

Dim1

HARMANDCONSEQUENCE_SA_ALCOHOL_RELATD_SOCL_CNSQNCE	92
HARMANDCONSEQUENCE_SA_ALCOHOL_RELATD_HLTH_CNSQNCE	92
HARMANDCONSEQUENCE_SA_ALCOHOL_RELATED_HARM	91

Name: count, dtype: int64

BACGROUP:

Dim1

BACGROUP_SA_GENERAL_POPULATION	348
BACGROUP_SA_YOUNG_NOVICE_DRIVERS	348
BACGROUP_SA_PROFESSIONAL_COMMERCE_DRIVERS	348

Name: count, dtype: int64

NATIONALSYSTEMTYPE:

NATIONALSYSTEMTYPE_SA_NST_SOCIALCONSEQUENCES	257
NATIONALSYSTEMTYPE_SA_NST_ALCOHOLPOLICYRESPONSES	218
NATIONALSYSTEMTYPE_SA_NST_HEALTHCONSEQUENCES	208
NATIONALSYSTEMTYPE_SA_NST_ALCOHOLCONSUMPTION	195

Name: count, dtype: int64

POLICYFRAMEWORK:

POLICYFRAMEWORK_INTEGRATED_SUBSTANCE_ABUSE	85
POLICYFRAMEWORK_INTEGRATED_GEN_PUB_HEALTH	85
POLICYFRAMEWORK_INTEGRATED_MENTAL_HEALTH	84
POLICYFRAMEWORK_INTEGRATED_OTHER	82
POLICYFRAMEWORK_INTEGRATED_NONCOM_DISEASES	81
POLICYFRAMEWORK_SPECIFIC_ALCOHOL_POLICY	81

Name: count, dtype: int64

PUBLICPLACE:

PUBLICPLACE_SA_LEISURE_EVENTS	260
PUBLICPLACE_SA_HEALTH_CARE_ESTABLISHMENTS	260
PUBLICPLACE_SA_GOVERNMENT_OFFICES	260
PUBLICPLACE_SA_PARKS_AND_STREETS	260
PUBLICPLACE_SA_EDUCATIONAL_BUILDINGS	260
PUBLICPLACE_SA_SPORTING_EVENTS	258
PUBLICPLACE_SA_PUBLIC_TRANSPORT	256
PUBLICPLACE_SA_RELIGIOUS_PLACES	249
PUBLICPLACE_SA_WORKPLACES	241

Name: count, dtype: int64

COORDINATINGENTITIES:

COORDINATINGENTITIES_OTHER_COORDINATING_ENTITY	86
COORDINATINGENTITIES_AGRICULTURE	85
COORDINATINGENTITIES_FINANCE_TAXATION	85
COORDINATINGENTITIES_HEALTH	84
COORDINATINGENTITIES_EDUCATION	83
COORDINATINGENTITIES_ROAD_SAFETY	83

COORDINATINGENTITIES_CRIMINAL_JUSTICE	83
COORDINATINGENTITIES_TRADE_INDUSTRY	82
COORDINATINGENTITIES_SOCIAL_AFFAIRS	82
COORDINATINGENTITIES_EMPLOYMENT	82

Name: count, dtype: int64

POLICYSECTOR:

Dim1

POLICYSECTOR_TRADE_INDUSTRY	121
POLICYSECTOR_EDUCATION	115
POLICYSECTOR_HEALTH_SECTOR	115
POLICYSECTOR_FINANCE_TAXATION	115
POLICYSECTOR_EMPLOYMENT	115
POLICYSECTOR_LAW_ENFORCEMENT	114
POLICYSECTOR_AGRICULTURE	111
POLICYSECTOR_JUSTICE	110
POLICYSECTOR_SOCIAL	109
POLICYSECTOR_ROAD_SAFETY	109

Name: count, dtype: int64

ENVCAUSE:

Dim1

ENVCAUSE_ENVCAUSE001	388
ENVCAUSE_ENVCAUSE000	388
ENVCAUSE_ENVCAUSE151	388
ENVCAUSE_ENVCAUSE060	388

Name: count, dtype: int64

POLICYADOPTIONLEVEL:

Dim1

POLICYADOPTIONLEVEL_OTHER_ADOPTION_LEVEL	16
POLICYADOPTIONLEVEL_SPECIFIC_MINISTRY	11
POLICYADOPTIONLEVEL_NATIONAL_GOVERNMENT	10
POLICYADOPTIONLEVEL_NATIONAL_PARLIAMENT	10

Name: count, dtype: int64

=====

DIM2 TYPES AND THEIR VALUES

=====

ASSISTIVETECHSOURCE:

Dim2

ASSISTIVETECHSOURCE_ASSISTIVETECH_Private	52
ASSISTIVETECHSOURCE_ASSISTIVETECH_NGO	52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Self-made	52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Other	52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Public	52
ASSISTIVETECHSOURCE_ASSISTIVETECH_Friends_and_family	52

Name: count, dtype: int64

AGEGROUP:

Dim2	
AGEGROUP_YEARS18-PLUS	305859
AGEGROUP_AGEAll	136244
AGEGROUP_YEARS10-19	113724
AGEGROUP_YEARS30-49	89661
AGEGROUP_YEARS15-29	89661
AGEGROUP_YEARS50-69	89112
AGEGROUP_YEARS70+	89112
AGEGROUP_YEARS05-14	89112
AGEGROUP_YEARS05-09	88326
AGEGROUP_YEARS05-19	88326

Name: count, dtype: int64

EMFFREQUENCY:

Dim2	
EMFFREQUENCY_EMFLOW	183
EMFFREQUENCY_EMFSTATIC	49

Name: count, dtype: int64

POP_TYPE:

Dim2	
POP_TYPE_POP_TYPE_HOUSEHOLD_HEAD	5864
POP_TYPE_CHILDREN	397
POP_TYPE_GENERALPOPULATION	397

Name: count, dtype: int64

PRICE:

Dim2	
PRICE_PRICE_IN_USD	3120
PRICE_CURRENCY	3120
PRICE_PRICE_IN_PPP	3120
PRICE_PRICE_IN_CURRENCY	3120
PRICE_QUANTITY	195

Name: count, dtype: int64

HOUSEHOLDCOOKINGFUEL:

Dim2

HOUSEHOLDCOOKINGFUEL_FUEL_KEROSENE	34068
HOUSEHOLDCOOKINGFUEL_FUEL_GAS	34068
HOUSEHOLDCOOKINGFUEL_FUEL_COAL	34068
HOUSEHOLDCOOKINGFUEL_FUEL_ELECTRICITY	34068
HOUSEHOLDCOOKINGFUEL_FUEL_CHARCOAL	34068
HOUSEHOLDCOOKINGFUEL_FUEL_BIOMASS	34068

Name: count, dtype: int64

BEVERAGE:

Dim2

BEVERAGE_BEV_BEER	975
BEVERAGE_BEV_SPIRITS	975
BEVERAGE_BEV_SUGAR_CARBONATED	390

Name: count, dtype: int64

TOBACCO_NICOTINE_PRODUCT:

Dim2

TOBACCO_NICOTINE_PRODUCT_CIGARETTE_MOSTSOLD	19695
TOBACCO_NICOTINE_PRODUCT_TOBACCO_OTHERSMOKED_MOSTSOLD	2535
TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_DISPOSABLE_CHEAPEST	2340
TOBACCO_NICOTINE_PRODUCT_ELILQUID_OPEN_CHEAPEST	2340
TOBACCO_NICOTINE_PRODUCT_TOBACCO_SMOKELESS_MOSTSOLD	2340
TOBACCO_NICOTINE_PRODUCT_ELILQUID_CLOSED_RECHARGABLE_CHEAPEST	2340
TOBACCO_NICOTINE_PRODUCT_TOBACCO_HEATED_HTP_MOSTSOLD	2145
TOBACCO_NICOTINE_PRODUCT_CIGARETTE_PREMIUM	1560
TOBACCO_NICOTINE_PRODUCT_CIGARETTE_CHEAPEST	1560

Name: count, dtype: int64

ENVCAUSE:

Dim2

ENVCAUSE_ENVCAUSE039	2727
ENVCAUSE_ENVCAUSE000	2727
ENVCAUSE_ENVCAUSE113	2727
ENVCAUSE_ENVCAUSE114	2727
ENVCAUSE_ENVCAUSE118	2727
ENVCAUSE_ENVCAUSE068	2727
ENVCAUSE_ENVCAUSE104	360

Name: count, dtype: int64

GHECAUSE:

Dim2

```

GHECAUSE_GHE001180    22620
GHECAUSE_GHE000390    22620
GHECAUSE_GHE001130    22620
GHECAUSE_GHE000680    22620
GHECAUSE_GHE000000    22620
GHECAUSE_GHE001140    22620
Name: count, dtype: int64
-----

```

```

ARCHIVE:
Dim2
ARCHIVE_DEC2019      16652
Name: count, dtype: int64
-----

```

```

TB_CASETYPE:
Dim2
TB_CASETYPE_TB_CASETYPE_BC    207
TB_CASETYPE_TB_CASETYPE_SP    190
Name: count, dtype: int64
-----

```

```

[57]: # Define aggregate/total values for each dimension type
aggregate_filters = {
    # Dim1 aggregates
    'SEX': ['SEX_BTSX'], # Both sexes
    'RESIDENCEAREATYPE': ['RESIDENCEAREATYPE_TOTL'], # Total (urban + rural)
    'AGEGROUP': ['AGEGROUP_YEARSALL'], # All ages
    'WEALTHQUINTILE': ['WEALTHQUINTILE_TOTL'], # Total wealth
    'EDUCATIONLEVEL': ['EDUCATIONLEVEL_TOTL'], # Total education
    'SEVERITY': ['SEVERITY_TOTAL'], # Total severity
    'HOUSEHOLDWEALTH': ['HOUSEHOLDWEALTH_TOTL'], # Total wealth
    'WEALTHTERCILE': ['WEALTHTERCILE_TOTL'], # Total wealth
    'ALCOHOLTYPE': ['ALCOHOLTYPE_SA_TOTAL'], # Total alcohol
    'TB_TREATMENTTYPE': ['TB_TREATMENTTYPE_TB_TREATMENTTYPE_TOTL'], # Total TB_
    ↪treatment
    'CONSUMPTIONTYPE': ['CONSUMPTIONTYPE_CONSUMPTION_TOTAL'], # Total_
    ↪consumption
    'ASSISTIVETECHPRODUCT': ['ASSISTIVETECHPRODUCT_ASSISTIVETECH_TOTAL'], #_
    ↪Total assistive tech

    # Dim2 aggregates
    'GHECAUSE': ['GHECAUSE_GHE000000'], # All causes
    'ENVCAUSE': ['ENVCAUSE_ENVCAUSE000'], # All environmental causes (likely)
}

# Also handle AGEGROUP in Dim2

```

```

dim2_agegroup_aggregates = ['AGEGROUP_AGEAll', 'AGEGROUP_YEARS18-PLUS']

df_filtered = df.copy()

# Filter Dim1 to aggregates
for dim_type, allowed_values in aggregate_filters.items():
    df_filtered = df_filtered[
        (df_filtered['Dim1Type'] != dim_type) | # Keep if not this dimension
        ↪type
        (df_filtered['Dim1'].isin(allowed_values)) # Or keep if it's an
        ↪aggregate value
    ]

# Filter Dim2 to aggregates
for dim_type, allowed_values in aggregate_filters.items():
    if dim_type == 'AGEGROUP':
        # Special handling for AGEGROUP in Dim2
        df_filtered = df_filtered[
            (df_filtered['Dim2Type'] != dim_type) |
            (df_filtered['Dim2'].isin(dim2_agegroup_aggregates))
        ]
    else:
        df_filtered = df_filtered[
            (df_filtered['Dim2Type'] != dim_type) |
            (df_filtered['Dim2'].isin(allowed_values))
        ]

# Count countries per indicator
indicator_coverage = df_filtered.groupby(['IndicatorCode',
'IndicatorName'])['SpatialDim'].nunique().reset_index(name='Country_Count')
indicator_coverage = indicator_coverage.sort_values('Country_Count',
    ↪ascending=False)

print(f"Total records before filtering: {len(df):,}")
print(f"Total records after filtering: {len(df_filtered):,}")
print(f"Reduction: {((len(df) - len(df_filtered)) / len(df) * 100):.1f}%")
print(f"\nNumber of unique indicators: {indicator_coverage.shape[0]}")
print(f"\nTop 200 indicators by country coverage:")

# Remove BMI indicators and add only NCD_BMI_30A indicator
top200Indicators = indicator_coverage[~indicator_coverage['IndicatorCode'].str.
    ↪lower().str.contains('bmi')].head(200)
top200Indicators = pd.
    ↪concat([indicator_coverage[indicator_coverage['IndicatorCode'] ==
    ↪'NCD_BMI_30A'], top200Indicators]).reset_index(drop=True)

```

```
display(top200Indicators)
```

```
top200Indicators[['IndicatorCode', 'IndicatorName']].to_csv('../data/metadata/
↳who_indicators_obesity_core.csv', index=False)
```

Total records before filtering: 3,961,866

Total records after filtering: 1,470,590

Reduction: 62.9%

Number of unique indicators: 924

Top 200 indicators by country coverage:

	IndicatorCode \		IndicatorName	Country_Count
0	NCD_BMI_30A		Prevalence of obesity among adults, BMI &Great...	210
1	NUTRITION_ANAEMIA_CHILDREN_PREV		Prevalence of anaemia in children aged 6-59 mo...	242
2	SDGIHR		Average of 13 International Health Regulations...	241
3	SA_0000001688		Alcohol, total per capita (15+) consumption (i...	235
4	WHS3_46		Total tetanus - number of reported cases	224
..	...		...	...
196	R_admin_duty_free_banned		Tobacco tax administration: Sales of duty (or ...	195
197	R_admin_identifier_track_and_trace		Tobacco tax administration: Marking/unique ide...	195
198	R_admin_overt_marking		Tobacco tax administration: Marking is overt	195
199	TOBACCO_MPOWER_M1_TOBACCOUSESMOKING		Tobacco Monitor: M1 population-based national ...	195
200	TOBACCO_MPOWER_M2_SMOKELESSECIGARETTES		Tobacco Monitor: M2 population-based national ...	195

[201 rows x 3 columns]

```
[64]: cols = ['IndicatorCode', 'IndicatorName',
             'SpatialDim', # Country code
             'SpatialDimType',
             'TimeDim', # Year
             'Dim1Type', 'Dim1', # (e.g., SEX)
             'Dim2Type', 'Dim2', # (e.g., AGEGROUP)
             'NumericValue', 'Value', # The actual data
```

```

        'Low', 'High' # Confidence intervals
    ]

df_tidy = df.copy()[cols]

#filter columns
df_tidy = df[[
    'IndicatorCode', 'IndicatorName',
    'SpatialDim', 'TimeDim',
    'Dim1Type', 'Dim1',
    'Dim2Type', 'Dim2',
    'NumericValue', 'Value',
    'Low', 'High'
]].copy()

# Rename for clarity
df_tidy = df_tidy.rename(columns={
    'SpatialDim': 'Country',
    'TimeDim': 'Year',
    'NumericValue': 'Numeric_Value',
    'Value': 'Text_Value'
})

# Create a unified value column (choose numeric when available)
df_tidy['Measurement'] = df_tidy['Numeric_Value'].fillna(df_tidy['Text_Value'])

# Separate by Sex if Dim1Type is SEX
df_tidy['Sex'] = df_tidy.apply(
    lambda x: x['Dim1'] if x['Dim1Type'] == 'SEX' else 'Both',
    axis=1
)

# Separate by Age if Dim2Type is AGEGROUP
df_tidy['AgeGroup'] = df_tidy.apply(
    lambda x: x['Dim2'] if x['Dim2Type'] == 'AGEGROUP' else 'All',
    axis=1
)

# Drop the generic Dim columns
df_tidy = df_tidy.drop(columns=['Dim1Type', 'Dim1', 'Dim2Type', 'Dim2'])

# Final tidy structure for 2012
df_tidy = df_tidy[df_tidy['Year'] == 2012][[
    'IndicatorCode', 'IndicatorName',
    'Country', 'Year',
    'Sex', 'AgeGroup',
    'Measurement', 'Low', 'High'
]]

```

```

]]

# Find indicators with strong correlation to NCD_BMI_30A

# Filter to numeric measurements only
df_analysis = df_tidy[pd.to_numeric(df_tidy['Measurement'],
errors='coerce').notna()].copy()
df_analysis['Measurement'] = pd.to_numeric(df_analysis['Measurement'])

# Create unique key for matching (Country + Year) - our focus is measurement_
↳per country and year
df_analysis['key'] = df_analysis['Country'] + '_' + df_analysis['Year'].
↳astype(str)

# Pivot: Make indicators as columns, keep country-year as rows (Sort by highest_
↳measurement before pivoting)
df_analysis = df_analysis.sort_values('Measurement', ascending=False)
df_pivot = df_analysis.pivot_table(
    index='key',
    columns='IndicatorCode',
    values='Measurement',
    aggfunc='first' # Use first value if duplicates
)

print(f"Pivot shape: {df_pivot.shape}")
print(f"Number of country-year observations: {df_pivot.shape[0]}")
print(f"Number of indicators: {df_pivot.shape[1]}")

# Only calculate for indicators with sufficient data
min_observations = 180

correlations = []
bmi_data = df_pivot['NCD_BMI_30A']

for col in df_pivot.columns:
    if ('BMI' in col.upper()) : #skip BMI indicators
        continue

    # Count valid pairs
    valid_pairs = bmi_data.notna() & df_pivot[col].notna()
    n_pairs = valid_pairs.sum()

    if n_pairs >= min_observations:

```

```

    corr = bmi_data.corr(df_pivot[col])
    correlations.append({
        'IndicatorCode': col,
        'Correlation': corr,
        'Correlation-abs': abs(corr),
        'N_observations': n_pairs
    })

# Sort by correlation
corr_df = pd.DataFrame(correlations).sort_values('Correlation-abs',
↪ascending=False)

# Filter for strong correlations (|correlation| > 0.4)
strong_corr = corr_df[corr_df['Correlation-abs'] > 0.4].copy()

print(f"\n Found {len(strong_corr)} indicators with |correlation| > 0.4")
print(f"    (from {len(corr_df)} total indicators with sufficient data)")

# 8. Add indicator names
indicator_names = df_tidy[['IndicatorCode', 'IndicatorName']].drop_duplicates()
strong_corr = strong_corr.merge(indicator_names, on='IndicatorCode', how='left')

# Reorder columns
strong_corr = strong_corr[['IndicatorCode', 'IndicatorName', 'Correlation',
    'N_observations']]

print("\n Strongly correlated indicators (sorted by correlation):")
display(strong_corr)

strong_corr.to_csv('../data/metadata/indicators_correlated_with_obesity.
↪csv', index=False)
print(f"\n Saved to: data/metadata/indicators_correlated_with_obesity.csv")

```

Pivot shape: (345, 249)

Number of country-year observations: 345

Number of indicators: 249

Found 14 indicators with |correlation| > 0.4
 (from 97 total indicators with sufficient data)

Strongly correlated indicators (sorted by correlation):

	IndicatorCode \
0	NCD_GLUC_03
1	NCD_GLUC_04
2	NCD_DIABETES_PREVALENCE_CRUDE


```

3  NCD_DIABETES_PREVALENCE_AGESTD
4      NCD_GLUC_01
5      NCD_PAA
6      NCD_PAC
7      TOTENV_2
8  PHE_HHAIR_PROP_POP_CLEAN_FUELS
9      TOTENV_4
10     WSH_SANITATION_BASIC
11         MCV2
12         AIR_52
13     WSH_WATER_BASIC

```

	IndicatorName	Correlation \
0	Raised fasting blood glucose (≥ 7.0 mmol/L) (...)	0.842245
1	Raised fasting blood glucose (≥ 7.0 mmol/L) (a...	0.830862
2	Prevalence of diabetes, crude	0.760636
3	Prevalence of diabetes, age-standardized	0.732298
4	Mean fasting blood glucose (mmol/l) (age-stand...	0.667667
5	Prevalence of insufficient physical activity a...	0.491062
6	Prevalence of insufficient physical activity a...	0.487815
7	Age-standardized deaths attributable to the en...	-0.477589
8	Proportion of population with primary reliance...	0.471137
9	Age-standardized DALYs attributable to the env...	-0.458480
10	Population using at least basic sanitation ser...	0.443851
11	Measles-containing-vaccine second-dose (MCV2) ...	0.417403
12	Household air pollution attributable death rat...	-0.412561
13	Population using at least basic drinking-water...	0.411288

	N_observations
0	201
1	201
2	210
3	210
4	192
5	205
6	205
7	192
8	196
9	192
10	200
11	203
12	187
13	200

Saved to: data/metadata/indicators_correlated_with_obesity.csv